MUSTER

Detect before you construct.

Use source structure to configure retrieval and context assembly.

01 USE THE SOURCE

Stored text as evidence

1. Alice's employer is Zynga
2. Alice's employer is Google

Blank-line records / messages
Continuous prose

No generative rewrite

→

02 DETECT

serial

Integer-dot share of non-empty lines.
≥0.80 → numbered branch.

recshare

Token-weighted bounded-record share.
≥0.90: records; otherwise: prose.
Protocol sets the reference scale.

No query, gold label or LLM in this decision.

→

03 CONFIGURE

Choose a retrieval plan

Unit: record or text chunk
Parent: slot or container
Order: serial, only where derived
Policy: parent-priority or balanced floors

04 RETRIEVE + RERANK

Dense + BM25 + parent graph
One shared cross-encoder score
Neighbour candidates only where configured

→

05 ASSEMBLE THE CONTEXT

Stored order; parent containers where configured.
Keep newest only when order/slot checks pass.
Keep the persistent store unchanged.

→

06 ANSWER FROM EVIDENCE

The reader uses the selected evidence.
Also use it on model-written memories,
without rebuilding the persistent store.

THE ABLATION: STRUCTURE CHANGES THE RESULT

MemoryAgentBench Overall · GPT-4o-mini reader · reported per-question budget

50.3

Hybrid retrieval + assembly
benchmark chunks, detection off

→

56.0

+ detected units, parents, order
balanced floors held fixed

→

57.5

+ structure-dependent policy gate
MUSTER

+5.7 points from detected representation; +1.5 further points from the policy gate.

Scope: the numbered branch recognizes integer-dot records; arbitrary dates or JSON IDs do not activate it.

Sources: MUSTER §3, Algorithm 1 and Table 2. LongMemEval keeps its native rounds / sessions as the documented exception.